DENIS KAHUTHU

Bristol, United Kingdom | kibuchikahuthu@gmail.com | +44 7523 153384 | github.com/Bucci-beep

PROFILE

Postgraduate researcher in Artificial Intelligence with an Information Technology foundation and practical expertise in predictive modeling, regression analysis, deep neural networks, and applied computational engineering. Experienced in designing rigorous Python-based ML experimentation pipelines, statistical hypothesis testing, uncertainty quantification, and error analysis across complex structured and spatial datasets. Proven track record of developing reproducible machine learning architectures, automated validation suites, and deployable intelligent prediction systems tailored for data-driven scientific discovery.

TECHNICAL SKILLS

Machine Learning & Predictive Modeling: Regression analysis (Linear, Logistic, Ridge/Lasso), classification algorithms, Random Forests, Support Vector Machines (SVM), cross-validation, hyperparameter optimization, Principal Component Analysis (PCA), feature engineering, model calibration.

Deep Learning & Neural Networks: Convolutional Neural Networks (CNNs), DenseNet121, transfer learning, 1D bio signal feature extraction, deep surrogate modeling, GradCAM explainability, out-of-distribution (OOD) detection, uncertainty estimation.

Programming, Libraries & Frameworks: Python, PyTorch, scikit-learn, Pandas, NumPy, SciPy, R, SQL, PostgreSQL, FastAPI, Pydantic, Git, Bash.

Computational Methods & Validation: Statistical inference, residual diagnostics, hypothesis testing (Shapiro-Wilk, McNemar, independent t-test), automated test suites (pytest), reproducible research reporting.

EDUCATION

University of the West of England (UWE Bristol) Bristol, UK | Sept 2025 – Sept 2026 Master of Science (MSc) in Artificial Intelligence

Relevant Modules: Machine Learning Algorithms, Statistical Inference, AI for Search and Optimization, Machine Learning for Language and Vision, Big Data, Cloud Computing, Data and Information Governance.

Technical University of Mombasa Mombasa, Kenya | 2016 – 2021 Bachelor of Technology (BTech) in Information Technology

Core Study: Advanced Programming, Applied Mathematics, Relational Databases, Algorithms & Data Structures, Systems Analysis, Computer Networks.

RELEVANT RESEARCH & COMPUTATIONAL PROJECTS

Tuberculosis Image Classification Pipeline MSc Machine Learning Project Deep Convolutional Neural Networks & Interpretable Baseline Modeling

- Engineered an end-to-end predictive computer vision pipeline across 19,938 image patches, systematically progressing from a regularized Logistic Regression baseline to custom CNNs and DenseNet121 deep architectures.
- Optimized feature representations via Principal Component Analysis (PCA), stratified k-fold cross-validation, and grid search, compressing the 4,096-dimensional input space while stabilizing gradient convergence. Conducted rigorous error analysis on false positive and negative distributions, isolating failure modes in flattened feature vectors to justify spatial inductive bias in deep convolutional filters.
- Achieved 92.48% test classification accuracy with DenseNet121, conducting formal McNemar statistical hypothesis testing to confirm statistically significant superiority over baseline models ($p < 0.01$).

Statistical Inference & Multiple Linear Regression Modeling MSc Quantitative Research Project Predictive Modeling, Assumption Diagnostics & Variance Analysis (R / Python)

- Formulated a multivariate predictive regression model analyzing observational cohort data from 237 subjects, addressing missing data imputation, outlier identification, and categorical feature encoding. Validated core regression assumptions through comprehensive diagnostic testing, including Breusch-Pagan homoscedasticity checks, Shapiro-Wilk residual normality tests, and Variance Inflation Factor (VIF) collinearity screening.
- Demonstrated strong predictive power with the final model explaining ~85% of observed variance ($R^2 = 0.85$), translating statistical parameters into actionable sensitivity and coefficient interpretations.

Agentic Bio-Signal Analysis & Intelligent Prediction System MSc Collaborative AI Research Project 1D-CNN Architectures, Federated Learning & Out-of-Distribution Controls

- Developed multi-format ingestion, signal preprocessing, and deterministic reasoning layers for physiological wave signals across 18,347 test instances.
- Implemented uncertainty estimation and out-of-distribution (OOD) routing mechanisms, demonstrating that headline accuracy (92.82%) masked class-imbalance failure modes (macro F1 of 0.455), prompting targeted sample weighting.
- Integrated GradCAM visual interpretability to trace neural network activations back to specific morphological wave intervals, providing interpretable feature attribution.
- Architected an automated verification suite containing 341 unit and regression tests, correcting signal interval rule anomalies against benchmark reference databases (MIT-BIH).

Knee MRI Volumetric Abnormality Pipeline Independent Computational Project (In Progress) Multi-Planar DICOM Ingestion, Spatial Structuring & Preprocessing

- Engineered a Python-based DICOM ingestion and metadata extraction engine for the MRNet volumetric dataset, physically ordering multi-slice imaging series across sagittal, coronal, and axial planes.
- Built automated validation algorithms to detect and isolate corrupted slices or mixed-series anomalies, ensuring structured spatial inputs for deep convolutional neural network feature extractors.

Asynchronous Distributed Multi-Agent AI System Independent Research Prototype Parallel Workflow Orchestration & Live Context Integration

- Designed a 5-agent asynchronous computational graph in Python utilizing asyncio and structured API integrations, cutting execution latency by over 50% compared to sequential execution models. Implemented real-time data ingestion, dynamic prompt routing, and error-recovery mechanisms to ensure robust multi-stage algorithmic convergence.

WORK EXPERIENCE

Fed Cafe Bristol, UK | Nov 2025 – Present Kitchen Porter / Operations Support

- Maintain rigorous operational standards and hygiene procedures within a fast-paced environment, coordinating with multidisciplinary team members under strict service deadlines.
- Adapt dynamically to changing daily workflow requirements, demonstrating dependable execution, meticulous attention to detail, and team coordination.